Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 6

Title: Sorting & Filtering

(Sales Performance Analysis Using Microsoft Power BI)

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Date: 26/10/2026

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1. Problem Statement

A company generates a large volume of sales data containing transaction details, products, categories, regions, revenue, profit, discounts, and returns. Analyzing this data manually makes it difficult for the sales manager to identify high-value transactions, compare regional sales performance, and recognize underperforming or loss-making products.

The objective is to use Microsoft Power BI to apply sorting and filtering techniques to organize sales data, identify important business patterns, compare revenue and profitability, and develop an interactive dashboard that supports effective business decision-making.

2. Objectives

The main objectives of this assignment are:

- To sort sales transactions according to revenue and profit.
- To identify high-value sales transactions.
- To filter sales data by date, region, category, and product.
- To identify underperforming and loss-making products.
- To compare revenue and profit across different regions and categories.
- To create an interactive Power BI dashboard for business decision-making.

3. Dataset Description

The dataset contains 300 sales transactions covering October 2025 to September 2026. It includes 12 products and four geographical regions.

The major fields used in the analysis are:

Field	Description
Transaction ID	Unique identifier for each transaction
Date	Date on which the transaction occurred
Region	Geographical sales region
Product	Name of the product sold
Category	Product category
Quantity	Number of units sold
Revenue	Total revenue generated
Profit	Profit earned from the transaction
Discount	Discount applied to the sale
Returned	Indicates whether the transaction was returned

4. Tools and Technologies Used

Microsoft Power BI Desktop was used to import, transform, analyze, and visualize the sales data. The following features were used:

- Power Query for data cleaning and transformation.
- DAX (Data Analysis Expressions) for calculating business metrics.
- Charts and tables for visualizing sales performance.
- Slicers for interactive filtering.
- Sorting and visual-level filters for identifying important transactions and products.

5. Methodology

Step 1: Importing the Data

I imported the Excel dataset into Power BI using the Get Data option. I selected the sales data sheet and checked the column names and data types to ensure the dataset was suitable for analysis.

Step 2: Data Preparation

I verified the date, numerical, and categorical columns. Revenue and profit were formatted as currency, while discount and profit margin were formatted as percentages.

I also created a calendar table to support chronological analysis of monthly sales.

Step 3: Creating Measures

I created DAX measures to calculate total revenue, total profit, transaction count, profit margin, and loss-making transactions.

For example:

```
Total Revenue =
SUM('Sales_Data'[Revenue ($)])
Total Profit =
SUM('Sales_Data'[Profit ($)])
Total Transactions =
DISTINCTCOUNT('Sales_Data'[Transaction ID])
Profit Margin % =
DIVIDE([Total Profit], [Total Revenue], 0)
```

These measures helped me evaluate business performance using numerical indicators.

6. Sorting Techniques

Sorting was used to arrange sales records in a meaningful order.

6.1 Sorting by Revenue

I sorted the Revenue column in descending order to identify the highest-value transactions first. This helped me understand which transactions contributed the most revenue to the business.

6.2 Sorting Products by Revenue

I sorted products from highest to lowest revenue using a horizontal bar chart. This made it easier to compare products and identify the strongest revenue contributors.

6.3 Sorting by Profit

I sorted products and transactions by profit to identify highly profitable items and transactions that generated losses.

6.4 Sorting Monthly Revenue

I organized monthly revenue chronologically to understand changes in sales over time. A calendar table and a year-month sorting column helped ensure the months appeared in the correct order.

7. Filtering Techniques

Filtering was used to display only records relevant to specific business questions.

7.1 Filtering High-Value Transactions

I applied a revenue filter to identify transactions with revenue greater than or equal to $32,400. This threshold was used as an example for the assignment.

7.2 Filtering by Region

I used a Region slicer to analyze sales in East, West, North, and South. This allowed me to compare geographical performance.

7.3 Filtering by Category and Product

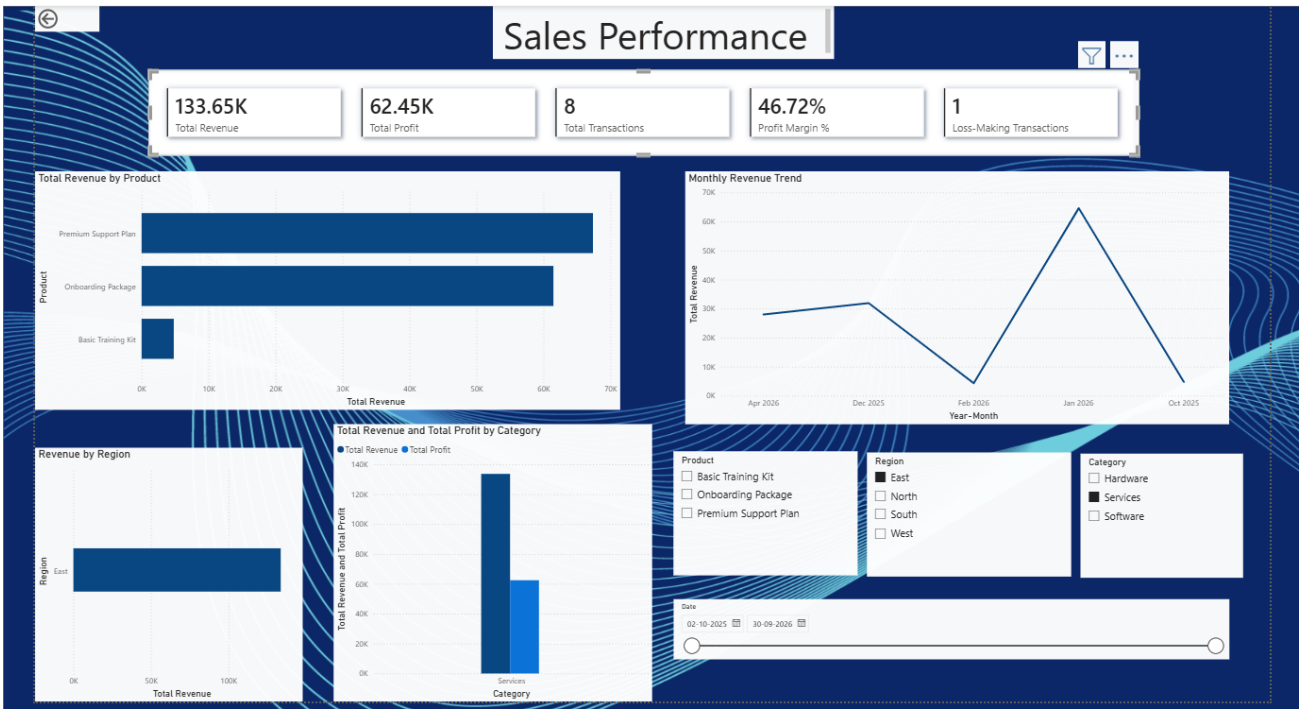
I used Category and Product slicers to analyze specific product groups and individual products.

7.4 Filtering by Date

A date-range slicer allowed me to select a particular period and analyze sales performance within that period.

7.5 Identifying Underperforming Products

I compared product revenue and profit margins to identify products requiring further investigation. Products with negative profit or low revenue relative to the average were flagged for review.



8. Power BI Dashboard

I created an interactive dashboard titled Sales Performance | Executive Overview.
The dashboard includes the following visualizations:
1. KPI Cards: Display total revenue, total profit, total transactions, profit margin, loss-making transactions, and return rate.
2. Monthly Revenue Trend: A line chart showing monthly revenue changes over time.
3. Revenue by Region: A horizontal bar chart comparing revenue across geographical regions.
4. Revenue and Profit by Category: A column chart comparing the financial performance of product categories.
5. Top Products by Revenue: A horizontal bar chart ranking the highest-revenue products.
6. Interactive Slicers: Date, region, category, and product slicers allow users to explore the data dynamically.
These visuals provide a consolidated view of sales performance and allow the manager to investigate specific business areas.

9. Results and Findings

Based on the full dataset analysis, the following findings were identified:

Metric	Result
Total Revenue	$4,289,922
Total Profit	$1,706,242
Total Transactions	300
Profit Margin	Approximately 39.77%
High-Value Transactions	32
Loss-Making Transactions	13

The high-value transaction count uses the threshold of $32,400.

Key Observations

1. Revenue and profitability: The business generated approximately $4.29 million in revenue and $1.71 million in profit during the analyzed period.
2. Regional performance: The West region generated the highest revenue in the full dataset, with $1,216,737.
3. Loss-making products: Smart Display 55in recorded a loss of $6,445, while Basic Training Kit recorded a loss of $4,750. These products require further investigation.
4. High-value sales: Sorting transactions by revenue made it easier to identify major sales and the products associated with them.
5. Interactive analysis: Slicers made it possible to examine specific regions, products, categories, and periods without manually searching through all transactions.

Note: Dashboard values can change when filters are applied. The figures above represent the full dataset rather than a filtered selection.

10. Business Recommendations

Based on the analysis, I recommend the following actions:

- Investigate the costs, pricing, and discounting strategies of loss-making products.
- Analyze the sales strategies used in the West region to understand its strong revenue contribution.
- Monitor high-value transactions to understand which products and customers generate significant revenue.
- Review return-related transactions to identify potential quality or customer-satisfaction issues.
- Use monthly sales trends to support inventory planning and sales forecasting.
- Regularly monitor product profitability rather than relying only on revenue.

11. Conclusion

In this assignment, I used Microsoft Power BI to apply sorting and filtering techniques to sales data. I created an interactive dashboard to examine revenue, profit, transaction values, regional performance, product performance, and monthly sales trends.

Sorting helped me rank transactions and products according to revenue and profit, while filtering enabled focused analysis by date, region, category, and product. The analysis also highlighted loss-making products and high-value transactions that require managerial attention.

Overall, this assignment helped me understand how sorting, filtering, DAX measures, and interactive visualizations can convert raw sales data into useful business insights and support better decision-making.